Course: Calculus II **Code:** MA1220 **Year:** 2025

Instructor: Dr. Neeraj Kumar **Department:** Department of Mathematics, IIT Hyderabad

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Calculus II (MA1220) - Problem Sheet 1

Q.1 Find the natural domains of the following functions of two variables:

(i) $\ln\left(9 - x^2 - 9y^2\right)$

(ii) $\sin^{-1}\left(x + y\right)$

(iii) $\dfrac{xy}{x^2 - y^2}$

(iv) $(x - 3y)^{\frac{1}{4}}$.

Q.2 Draw contour maps of the following functions, showing suitable level curves.

(i) $x - y$

(ii) $\sqrt{x} + y$

(iii) $x^2 - y^2$

(iv) xy

Q.3 Answer the following:

(i) If $f(x,y) = \dfrac{x^2y^2}{x^2y^2 + (x-y)^2}$ for $(x,y) \neq (0,0)$, does $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exist?

(ii) If $f(x,y) = \dfrac{x^3 + y^3}{x^2 + y^2}$ for $(x,y) \neq (0,0)$, does $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exist?

(iii) If $f(x,y) = (x^2 + y^2)\ln(x^2 + y^2)$ for $(x,y) \neq (0,0)$, does $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exist?

(iv) If $f(x,y,z) = \dfrac{xy + yz}{x^2 + y^2 + z^2}$ for $(x,y,z) \neq (0,0,0)$, does $\lim\limits_{(x,y,z)\to(0,0,0)} f(x,y,z)$ exist?

Theorem 0.1. *Let $f, g : \mathbb{R} \to \mathbb{R}$ be continuous. Define $F : \mathbb{R}^2 \to \mathbb{R}$ and $G : \mathbb{R}^2 \to \mathbb{R}$ by*

$$F(x,y) = f(x) \quad \text{and} \quad G(x,y) = g(y).$$

Then prove that the following functions are continuous on $\mathbb{R}^2$:

(i) $F(x,y) + G(x,y)$;

(ii) $F(x,y) - G(x,y)$;

(iii) $F(x,y)\,G(x,y)$;

(iv) $\max\{F(x,y), G(x,y)\}$;

(v) $\min\{F(x,y), G(x,y)\}$.

Q.4 Answer the following:

(i) Let

$$f(x,y) = \begin{cases} \dfrac{\sin(xy)}{xy} & \text{if } xy \neq 0, \\ 1 & \text{if } xy = 0. \end{cases}$$

Is f continuous on $\mathbb{R}^2$?

(ii) Let

$$f(x,y) = \begin{cases} (x^2+y^2)\,\sin\dfrac{1}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Is f continuous on $\mathbb{R}^2$?

(iii) Let

$$f(x,y) = \begin{cases} x\sin\dfrac{1}{x} \;+\; y\sin\dfrac{1}{y} & \text{if } x \neq 0 \text{ and } y \neq 0, \\ x\sin\dfrac{1}{x} & \text{if } x \neq 0 \text{ and } y = 0, \\ y\sin\dfrac{1}{y} & \text{if } y \neq 0 \text{ and } x = 0, \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Is f continuous on $\mathbb{R}^2$?

(iv) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x,y) = x + y$ for all $(x,y) \in \mathbb{R}^2$. Prove that f is continuous on $\mathbb{R}^2$.

(v) Let

$$f(x,y) = \begin{cases} \dfrac{\pi x^2 y}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Is f continuous on $\mathbb{R}^2$?

Q.5 Answer the following:

(i) Use the limit definition of partial derivatives to calculate f_x and f_y for each of the following functions:

$$f(x,y) = xy^2 - x^3y, \qquad f(x,y) = \frac{x}{x+y^2}.$$

(ii) The temperature at a point (x,y) on a flat metal plate is given by

$$T(x,y) = \frac{60}{1+x^2+y^2} \quad (\text{in } ^\circ\text{C}), \qquad x, y \text{ in meters.}$$

Find the rate of change of temperature with respect to distance at the point $(2,1)$ in (a) the x-direction and (b) the y-direction.

(iii) Let

$$f(x,y) = \begin{cases} (x^2+y^2)\,\sin\dfrac{1}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Do the partial derivatives of f exist at $(0,0)$?

(iv) Let

$$f(x,y) = \begin{cases} x\sin\dfrac{1}{x} \;+\; y\sin\dfrac{1}{y} & \text{if } x \neq 0 \text{ and } y \neq 0, \\ x\sin\dfrac{1}{x} & \text{if } x \neq 0 \text{ and } y = 0, \\ y\sin\dfrac{1}{y} & \text{if } y \neq 0 \text{ and } x = 0, \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Do the partial derivatives of f exist at $(0,0)$?

(v) Suppose you are told there exists a function $f(x,y)$ whose partial derivatives are

$$f_x(x,y) = x + 4y \quad \text{and} \quad f_y(x,y) = 3x - y.$$

Should you believe this?

Q.6 Find an equation of the tangent plane to the given surface at the specified point.

(i) $z = 2x^2 + y^2 - 5y$ at the point $(1, 2, -4)$.

(ii) $z = x\sin(x+y)$ at the point $(-1, 1, 0)$.

(iii) $z = e^{x-y}$ at the point $(2, 2, 1)$.

Q.7 Answer the following:

(i) Show that $f(x,y) = x^2 + y^2$ is differentiable for all $(x,y) \in \mathbb{R}^2$.

(ii) Given that f is differentiable function with

$$f(2,5) = 6, \qquad f_x(2,5) = 1, \qquad f_y(2,5) = -1,$$

use a linear approximation to estimate $f(2.2, 4.9)$.

Q.8 Find the directional derivative of the function at the given point in the direction of the vector $\mathbf{v}$

(i) $f(x,y) = e^x \sin y$ at the point $(0, \frac{\pi}{3})$, $\mathbf{v} = \langle -6, 8\rangle$.

(ii) $g(s,t) = s\sqrt{t}$ at the point $(2,4)$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j}$.

(iiii) $f(x,y,z) = x^2y + y^2z$ at the point $(1,2,3)$, $\mathbf{v} = \langle 1,1,1\rangle$.

Q.9 Analyze the following functions for local maxima, local minima and saddle points:

(i) $f(x, y) = x^2 + xy + y^2 + y$.

(ii) $f(x, y) = y \cos x$.

(iiii) $f(x, y) = y(e^x - 1)$.

(iv) $f(x, y) = x^3 - 3x + 3xy^2$.

Q.10 Answer the following:

(i) Show that $f(x, y) = x^2 + 4y^2 - 4xy + 2$ has an infinite number of critical points and that $D = 0$ at each one. Then show that f has a local (and absolute) minimum at each critical point.

(ii) Find the absolute maximum and the absolute minimum values of the function $f(x, y) = (x^2 - 4x) \cos y$ on the rectangle $\mathcal{R} = \{(x, y) \mid 1 \leqslant x \leqslant 3, -\frac{\pi}{4} \leqslant y \leqslant \frac{\pi}{4}\}$.

(iiii) Use Lagrange multipliers to find the extreme values of the function $f(x, y) = 3x + y$ subject to the given constraint $x^2 + y^2 = 10$.

(iv) Use Lagrange multipliers to find the extreme values of the function $f(x, y) = x + y + z$ subject to the given constraints $x^2 + z^2 = 2$ and $x + y = 1$.